

Speaker Notes — Read Alongside Your Slides

Print this and keep it next to your laptop during the presentation.

SLIDE 1 — Title & Research Aim

“Good morning. My dissertation is titled ‘A Machine Learning Approach to Real-Time Phishing URL Detection: A Comparative Analysis of Classification Models.’

My research aim: to find out whether machine learning models trained only on lexical URL features can achieve high accuracy and low False Positive Rates for real-time phishing detection.

In simple terms — can we tell if a URL is dangerous just by looking at the text of the link, without loading the page?”

→ *Transition: “Let me give you some context on why this matters.”*

SLIDE 2 — Background & Rationale

“Phishing is the number one cybercrime. The APWG reported over 4.7 million attacks in 2023. The FBI estimates \$10 billion in losses.

The traditional defence is blacklisting — but Sheng et al. found it takes 12 hours for a new phishing URL to be added. By then, victims are already compromised.

We need a proactive system that analyses URLs the moment a user clicks — in real time, without waiting for a list update.”

→ *Transition: “Now let me show you what the existing research says.”*

SLIDE 3 — Literature Overview & Gap

“Three generations of detection. Blacklists — accurate but reactive. Content-based — accurate but slow. Lexical — fast, but is it accurate enough?

Key studies: Ma et al. 2009 proved URL features have predictive power. Sahingoz et al. 2019 showed Random Forest outperforms linear models. Rao and Pais 2020 used deep learning but

did not report FPR.

My gap: no controlled comparison of linear vs non-linear classifiers using only structural lexical features on a large dataset, with explicit FPR reporting.”

→ *Transition: “So how did I go about addressing this gap?”*

SLIDE 4 — Methodology

“CRISP-DM framework — the industry standard — mapped to two-week Agile sprints.

Positivist philosophy. Deductive approach — I had a hypothesis from the literature and tested it.

Dataset: 641,053 URLs from Kaggle after cleaning. $80/20$ stratified split. 10 lexical features extracted from the URL string alone.

Three classifiers: Logistic Regression, Random Forest, Support Vector Machine. Same data, same features, same split — a fair, controlled experiment.”

→ *Transition: “Let me explain the specific features I engineered.”*

SLIDE 5 — The 10 Lexical Features

“All 10 features are extracted from the URL text alone — no network calls, no page loading. Extraction takes less than 1 millisecond per URL.

Top three by ANOVA F-test: dot_count at 31,994, path_length at 19,041, slash_count at 10,162. All ten are statistically significant at p less than 0.001.

Why they work: attackers manipulate URLs to look legitimate. Subdomain stacking adds dots. Obfuscation adds path depth. These structural patterns are what the model learns.”

→ *Transition: “Now for the results.”*

SLIDE 6 — Results

“Random Forest: 94.03% accuracy, F1-Score 0.9120, FPR 5.45%.

Linear models: 78-79% accuracy, recall only 50% — they miss half the malicious URLs.

Cross-validation confirms stability: RF scored 91.95% with standard deviation of just 0.17%.

The 15-percentage-point gap between RF and linear models is the key finding.”

→ *Transition: “What do these numbers actually mean?”*

SLIDE 7 — Why Random Forest Won

“The relationship between URL features and phishing is non-linear.

A URL with many dots is not necessarily malicious — `bbc.co.uk` has three dots. But many dots combined with a long path and high entropy is very suspicious.

Linear models try to find one straight line through the feature space. Random Forest builds 200 trees that each split on different features — it can model complex combinations that linear models simply cannot.”

→ *Transition: “Of course, no study is perfect.”*

SLIDE 8 — Limitations & Discussion

“Three limitations I acknowledge honestly. Temporal drift — phishing tactics evolve. Lexical features alone cannot catch perfect mimicry. And I only tested three algorithms.

But three contributions: the largest controlled lexical comparison, explicit FPR reporting, and a fully reproducible open-source pipeline on GitHub.

The limitations do not undermine the value — they define the scope.”

→ *Transition: “So to bring it all together...”*

SLIDE 9 — Conclusion

“Machine learning on lexical features can detect phishing in real time with 94% accuracy and acceptable FPR — but only with non-linear ensemble methods. Linear models are not sufficient.

Future work: character n-grams, temporal retraining, gradient boosting, browser extension deployment.

The practical implication: this can be a fast, privacy-preserving first layer of defence anywhere you need instant URL classification.”

→ *Transition: “And finally, let me show you the working system.”*

SLIDE 10 — Live Demo

“This is PhishGuard — the working prototype.

I will test three URLs:

- `google.com` → Safe, score $\frac{5}{100}$

- A suspicious URL → Medium risk, around $50/100$
- An IP-based URL → Malicious, score $100/100$

It uses the same 10 features and scoring logic from the research. Runs entirely in the browser — no server needed.”

If internet fails: “I have screenshots in Appendix F showing all three result types.”

TIMING GUIDE

Slide	Target time	Cumulative
1. Title	30 sec	0:30
2. Background	1:30	2:00
3. Literature	1:30	3:30
4. Methodology	1:30	5:00
5. Features	1:00	6:00
6. Results	1:30	7:30
7. Why RF	1:00	8:30
8. Limitations	1:00	9:30
9. Conclusion	1:00	10:30
10. Demo	2:00	12:30

Total: ~12 minutes (leaves buffer within the 20-minute limit)